Preserving lesion quantification in longitudinal whole-body PSMA PET/CT registration

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Abstract. **Purpose:** Registering follow-up to baseline PSMA PET/CT scans enables voxel-wise comparison of tumour burden, but the deformation can itself change the metabolic tumour volume (MTV) and total lesion glycolysis (TLG) of lesions. We study losses that preserve these biomarkers and keep bones rigid, and whether they belong in network training or in instance optimisation (IO).

Approach: We trained LapIRN networks with and without PET-preservation and bone-rigidity losses on the Learn2Reg 2026 PSMAReg data, applied the losses in training, IO, or both, and compared against tuned NiftyReg and ConvexAdam on 28 held-out patients, with patient-level statistics and a per-lesion analysis. [TODO: Rigidity penalties compared.]

Results: All our variants were more accurate than NiftyReg and ConvexAdam (CT Dice 75.6–79.2 % vs. 69.4 % and 61.7 %) without folding. With IO, the PET losses lowered the MTV change from 10.2 % to 6.3 % and the TLG change from 11.9 % to 7.2 % at a cost of 2.8 Dice points. Applying the losses only in IO was 1.7 Dice points more accurate than also training with them, with similar preservation. IO removed almost all volume change of its predicted lesion masks; the remaining change stemmed mainly from missed lesions. [TODO: Rigidity result.]

Conclusions: When IO is affordable, biomarker-preserving losses are best applied there, on a network trained for accuracy; otherwise, they should be part of training. Further preservation requires better lesion segmentation rather than stronger loss weights.

Keywords: deformable image registration, PSMA PET/CT, longitudinal imaging, biomarker preservation, instance optimization.

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1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT is now routinely used for staging prostate cancer, for planning and following radioligand therapy, and for assessing response.[1] Patients therefore accumulate a baseline examination and one or more follow-up examinations, and registering each follow-up to its baseline makes it possible to compare tumour burden voxel by voxel, to track individual lesions, and to accumulate dose.[2] For these uses, a registration must do two things at

once: it has to align the anatomy, and it must not itself change the PET-derived quantities that are compared across time points. Two such quantities are the metabolic tumour volume (MTV) and the total lesion glycolysis (TLG). A deformation that compresses or stretches a lesion changes both, so a change measured after registration can be an artefact of the registration rather than of the disease. The Learn2Reg 2026 PSMAReg challenge[3–5] formalises this setting: each follow-up scan is warped to its baseline by a dense displacement field, which is scored on organ overlap, boundary distance, folding, and the change of MTV and TLG caused by the warp.

Classical registration solves an optimisation problem per image pair;[6–8] this is accurate but slow for whole-body scans. Learning-based registration replaces the per-pair optimisation with one forward pass of a trained network,[9–11] and instance optimisation (IO) can refine the network output per pair at test time. The coarse-to-fine Laplacian pyramid network LapIRN[12] recovers the large deformations between whole-body scans while keeping the transformation smooth and invertible.

In a challenge submission,[13] we adapted LapIRN to PSMA PET/CT and added losses that preserve the PET biomarkers of lesions and keep bones rigid. That submission was developed and evaluated within the challenge protocol. Here, we study these losses in a controlled setting: a new patient-level split with a held-out test set, retrained networks, tuned baselines, and statistics at the patient level. We ask two questions:

1. Do the PET-preservation and bone-rigidity losses improve the complete registration pipeline, and what do they cost in alignment accuracy?

2. Do the losses have to be part of network training, or is it sufficient to use them in instance optimisation?

Our contributions are:

- a held-out, patient-level evaluation of PET-preservation and bone-rigidity losses for whole-body PSMA PET/CT registration, against tuned NiftyReg and ConvexAdam;
- a comparison of where the losses act, showing that applying them only in IO preserves as well as training with them and is more accurate, while training with them is the only option without IO;
- a per-lesion analysis of which lesions registration deforms, and an analysis showing that preservation in IO is limited by the lesions that the predicted masks miss, not by the loss weights;
- a comparison of rigidity penalties for low-resolution whole-body CT, including stencil leakage of finite-difference penalties at bone boundaries. [TODO: Check against the rigidity results.]

2 Methods

2.1 Data and Split

We used the training set of the PSMAReg challenge,[2,5] a longitudinal whole-body PSMA PET/CT cohort. Each scan consists of a CT and a PET image (in SUV), resampled to $2.7344 \times 2.7344 \times 3.27$ mm and cropped to $192 \times 192 \times 288$ voxels. For every scan, the challenge provides 117 CT organ labels (the TotalSegmentator[14] classes) and a PSMA lesion mask. Following the challenge, a follow-up scan is always the moving image and the corresponding baseline scan the fixed image.

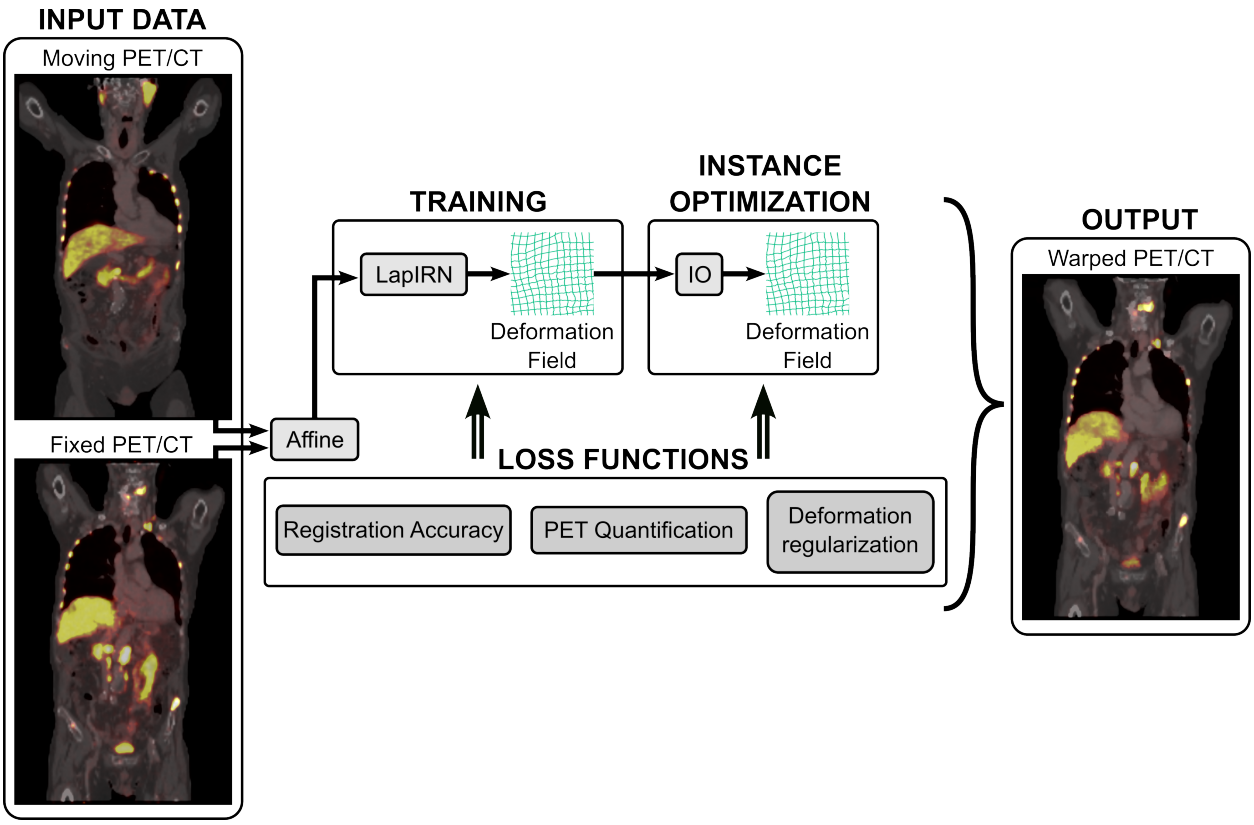


Fig 1 Registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT scans are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimisation (IO) refines this deformation per pair before the moving scan is resampled into the fixed frame. Network training and IO use the same groups of loss terms: alignment, PET preservation, and deformation regularisation (Sec. 2.3). [TODO: Update the figure if the rigidity term or the IO-only variant should appear.]

We split the 146 labelled patients with at least two scans at the patient level into training, validation, and test sets, aiming at 70/15/15 % of the registration pairs and balancing the sets for tumour burden and for bone- versus soft-tissue lesion location. Training uses all pairs of later onto earlier examinations of a patient; validation and testing use only follow-up-to-baseline pairs, as in the challenge. This gave 92/26/28 patients and 178/38/38 pairs. The split was built once with a fixed seed and frozen before any experiment of this study; all choices of settings, checkpoints, and methods were made on the validation set, and the test set was used only for the final evaluation. All patients were part of the challenge training data used to develop the earlier version of the method;[13] the test patients were not used for any decision in the present study.

Labels used by the method and for evaluation. Network training uses the provided organ and lesion labels of the training patients. Instance optimisation (Sec. 2.2) also needs organ and lesion labels, but of the scans being registered; to keep the evaluation independent of these inputs, IO uses only predicted labels, and the provided labels of validation and test scans are used only for evaluation (Sec. 3.1). Organ labels are predicted from the CT by TotalSegmentator.[14] Lesion masks are predicted from PET/CT by an nnU-Net[15] set up as the winning autoPET III solution:[16] a residual-encoder U-Net (ResEnc-L) with 192^3 patches, initialised from their pretrained weights, with an auxiliary organ-segmentation head and misalignment augmentation. It was trained with patient-grouped five-fold cross-validation on all 494 scans of the non-test patients, including patients with a single examination. Validation scans are segmented by the fold model that did not see the patient, and test scans by the ensemble of the five fold models. The predicted lesion masks reach a mean Dice of 0.56 on the validation scans and 0.51 on the test scans (medians 0.69 and 0.67).

Preprocessing. For the network and IO, CT was clipped to $[-1000, 1500]$ HU and PET to $[0, 20]$ SUV, and both were rescaled to $[0, 1]$. As in the ConvexAdam baseline of the challenge,[17] the patient bed was removed from the CT. All PET quantities in the evaluation are computed on the original SUV images.

2.2 Registration Pipeline

The pipeline (Fig. 1) has three stages, all estimated from the moving scan (C_m, P_m) and the fixed scan (C_f, P_f), where C denotes CT and P PET. We denote the affine transform by φ_{aff}, the network deformation by φ_θ, and their composition, which maps the original moving scan to the fixed scan, by φ.

Affine pre-registration. ANTs[18] estimates an affine transform from the two CTs, using body masks, a $[-1000, 1000]$ HU window, half-resolution images, and Mattes mutual information. It is converted to a dense displacement field.

Deformable network. We use LapIRN[12] with three pyramid levels at quarter, half, and full resolution. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level and integrated by scaling and squaring[19] into φ_θ. Each level receives four channels: the fixed CT and PET and the affinely aligned moving CT and PET. The levels are trained one after the other; a finer level is initialised from the coarser one, which is first kept frozen and then trained jointly. Each sub-network has 24 base channels and a trunk of ten residual blocks with an expansion factor of two.

Instance optimisation. At test time, the deformation of each pair is refined by optimising a residual stationary velocity field, initialised at zero and integrated by scaling and squaring (seven steps), on top of the total transformation φ. The residual is optimised at full resolution with Adam[20] (learning rate 0.0125, 40 iterations). The objective is the same as in training (Sec. 2.3), with the predicted labels in place of ground-truth labels. Of the 40 iterates, we return the one with the lowest objective, in which the interpolated (soft) MTV and TLG terms are replaced by their nearest-neighbour (hard) values, i.e. by how the evaluation counts them; this selection uses only predicted labels.

2.3 Registration Objective

Network training and IO minimise a weighted sum of three groups of terms:

$$\mathcal{L} = \underbrace{\lambda_{\mathrm{NCC}}\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\mathcal{L}_{\mathrm{DSC}}}_{\text{alignment}} + \underbrace{\lambda_{\mathrm{MTV}}\mathcal{L}_{\mathrm{MTV}} + \lambda_{\overline{J}}\mathcal{L}_{\overline{J}} + \lambda_{J}\mathcal{L}_{J} + \lambda_{\mathrm{TLG}}\mathcal{L}_{\mathrm{TLG}}}_{\text{PET preservation}} + \underbrace{\lambda_{\mathrm{rig}}\mathcal{L}_{\mathrm{rig}} + \lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\mathcal{L}_{\mathrm{NDV}}}_{\text{regularisation}}. \tag{1}$$

All terms are evaluated on the total transformation φ. The PET-preservation terms and the rigidity term are the subject of this study; they can be switched on in network training, in IO, or in both (Sec. 3). The two coarse network levels never use them, because lesions and thin bones span only a few voxels at these resolutions.

Alignment. $\mathcal{L}_{\mathrm{NCC}}$ is the negative local normalised cross-correlation of the CTs, summed over one to three scales depending on the pyramid level. PET is not used for similarity, because uptake changes between examinations with disease and therapy and a PET similarity would reward

matching these changes. $\mathcal{L}_{\mathrm{DSC}}$ is a soft Dice loss on the CT organ labels.

PET preservation. Let M be the lesion mask of the moving scan and $\widetilde{M}$, $\widetilde{P}$ the linearly warped mask and PET. Nearest-neighbour warping, as used in the evaluation, is piecewise constant in the displacement and gives no useful gradient; linear warping gives a soft lesion boundary instead. The volume and uptake terms penalise the relative change caused by warping,

$$\mathcal{L}_{\mathrm{MTV}} = \left(\frac{\left| \sum_x \widetilde{M}(x) - \sum_x M(x) \right|}{\sum_x M(x) + \epsilon} \right)^2, \qquad \mathcal{L}_{\mathrm{TLG}} = \frac{\left| \sum_x \widetilde{P}(x)\widetilde{M}(x) - \sum_x P_m(x)M(x) \right|}{\sum_x P_m(x)M(x) + \epsilon}, \tag{2}$$

with $\epsilon = 10^{-5}$. Two further terms act on the Jacobian determinant $\det J_\varphi$ inside the warped lesion region W in the fixed frame:

$$\mathcal{L}_{\overline{J}} = \left(\frac{1}{|W|} \sum_{x \in W} \det J_\varphi(x) - 1 \right)^2, \qquad \mathcal{L}_J = \frac{1}{|W|} \sum_{x \in W} \big(\det J_\varphi(x) - 1 \big)^2. \tag{3}$$

$\mathcal{L}_{\overline{J}}$ constrains the net volume change of all lesions, and $\mathcal{L}_J$ additionally prevents compression in one part of a lesion from being compensated by expansion in another. For $\mathcal{L}_{\overline{J}}$, W is the linearly warped mask (without gradient through the mask), for $\mathcal{L}_J$ the nearest-neighbour warped mask.

Bone rigidity. Many PSMA lesions lie in bone, which moves rigidly between examinations. For each of the 61 skeletal labels of the fixed scan (in IO: the predicted labels), we fit a rigid transformation to the displacement of the structure in closed form[21,22] and penalise the mean squared residual of its voxels; $\mathcal{L}_{\mathrm{rig}}$ is the mean over structures with at least 50 voxels. Each bone is fitted independently, so neighbouring bones may move relative to each other, and the term involves no finite differences, so it does not constrain the tissue around a bone.

Table 1 Loss weights of Eq. (1) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimisation (IO).

	λ_{NCC}	λ_{DSC}	λ_{MTV}	$\lambda_{\overline{J}}$	λ_{J}	λ_{TLG}	λ_{rig}	$\lambda_{\mathrm{smooth}}$	λ_{NDV}
Training	5	5	5	0.125	1.25	1.25	0.05	10	10^4
IO	5	5	$4 \cdot 10^4$	10^3	4	16	0.25	0	10

Regularisation. $\mathcal{L}_{\mathrm{smooth}}$ is the mean squared spatial gradient of the displacement. $\mathcal{L}_{\mathrm{NDV}}$ is a differentiable version of the challenge's non-diffeomorphic volume, which splits every voxel into tetrahedra and accumulates the negative part of their Jacobian determinants.[23]

Weights. Table 1 lists the weights. In training, the alignment weights are the same for all models, and the PET and rigidity weights are either used or set to zero. The IO weights were tuned on the validation set (Sec. 3.3).

2.4 Rigidity Penalties Compared

Besides the per-bone rigid fit, we evaluate the rigidity penalty of Staring et al.,[24] which requires the Jacobian J inside a bone to be a rotation: $\det J = 1$, $J^\top J = I$, and vanishing second derivatives of the deformation. It is computed with finite differences, whose stencil reads the 6 face and 12 edge neighbours of a voxel. At a bone boundary, part of the stencil lies outside the bone, so the penalty also acts on the surrounding soft tissue or on a neighbouring bone. A common remedy is to erode the bone mask so that the stencil stays inside; we erode each bone by exactly this 18-neighbour footprint, which removes the leakage completely. The three variants (Staring, Staring with erosion, per-bone rigid fit) are compared in IO (Sec. 3).

2.5 *Training*

Each level was trained with Adam[20] for 10^5, 10^5, and $1.2 \cdot 10^5$ steps (quarter, half, full resolution) with learning rates of $1.5 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, and 10^{-5}, a batch size of one with gradients accumulated over four pairs, a linear warm-up over five epochs, and a cosine decay to 10 % of the initial rate. A finer level is trained jointly with the coarser ones after ten epochs. Convolutions run in bfloat16, all losses in single precision. Training took 18 h, 22 h, and 3.7 days for the three levels on one NVIDIA A100 (80 GB). The augmentations mimic longitudinal variability: left–right flipping ($p = 0.5$), CT intensity shifts of ± 0.02 in the normalised window (about ± 50 HU), CT scaling between 0.9 and 1.1, PET scaling between 0.85 and 1.15, and field-of-view mismatch by cropping up to 40 axial slices from both scans and up to 10 further slices from each scan separately. The two coarse levels are shared by all models; the variants of this study differ only in the losses of the full-resolution level. Checkpoints were selected on the validation set: the coarse levels by CT Dice, the full-resolution level by the best accuracy (CT Dice and HD95) among checkpoints whose validation folding stays below $1.4 \cdot 10^{-4}\,\%$ non-diffeomorphic volume. We did not select on MTV or TLG, so that the PET losses are not credited with effects of the checkpoint choice.

Use of AI tools. Parts of the code for training, evaluation, statistical analysis, and figure generation were written with the help of an AI coding assistant. All generated code was reviewed and tested by the authors, who are responsible for the analyses and results.

3 Experiments

3.1 Evaluation Metrics

All metrics are computed with the provided labels of the test scans, after warping the moving scan with the scorer of the challenge.

Alignment. CT Dice and the 95th percentile of the Hausdorff distance (HD95), averaged over the 117 organ labels.

PET preservation. As in the challenge, PET (SUV) is warped linearly and the lesion mask by nearest neighbour. MTV is the volume of the lesion mask and TLG the sum of SUV over it times the voxel volume; both are measured on the moving scan before and after warping. The error of $Q \in \{\mathrm{MTV}, \mathrm{TLG}\}$ is the absolute relative change,

$$|\Delta Q| = \frac{|Q_{\mathrm{after}} - Q_{\mathrm{before}}|}{\max(Q_{\mathrm{before}}, 10^{-6})}. \tag{4}$$

This measures what the registration does to the same scan; it does not assume that disease is unchanged between examinations. Because PET and mask are warped separately, the scorer changes TLG even for the identity transformation: on the test pairs with lesions, an all-zero displacement changes TLG by $2.9 \pm 1.2\,\%$ (MTV: $0\,\%$). This resampling floor applies to every warped method; we therefore report the unregistered scans ("Initial") without resampling.

Per-lesion change. The aggregate errors of Eq. (4) can hide large changes of individual lesions that cancel. We therefore also split the moving lesion mask into 26-connected components and compute $|\Delta\mathrm{MTV}|$ for each. Lesions are grouped by size (small < 0.5 ml, medium < 5 ml, large)

and location (bone-associated if at least half of the lesion lies on a skeletal label); lesions with fewer than five voxels are excluded.

Deformation quality. Percentage of non-diffeomorphic volume (NDV) inside the body, as defined by the challenge.[23]

Bone metrics. For the rigidity comparison, we additionally compute Dice and HD95 separately over bones and soft-tissue organs; the distortion of within-bone distances, i.e. the mean relative change of the physical distance between randomly sampled pairs of interior points (eroded mask) of the same bone, separately for short (5–15 mm) and long ($\geq$ 15 mm) distances; NDV inside bones and in a three-voxel band around them; and, as a secondary measure, the residual of a rigid fit to each bone's displacement (bone rigidity error). We also measure the fraction of stencil reads that leave the bone of the centre voxel (stencil leakage) and the fraction of bone voxels and of bones that remain after erosion.

3.2 Statistical Analysis

A patient may contribute several pairs and many lesions. We therefore average every metric over the pairs of a patient first and use patients as the unit of all tests, rankings, and confidence intervals (28 test patients). Paired differences between two methods are tested with the two-sided Wilcoxon signed-rank test[25] on patient values and corrected for multiple comparisons with Holm's method[26] within each table and metric; we call a difference significant if the corrected $p < 0.05$. Confidence intervals are 95 % percentile intervals over patient bootstrap samples;[27] for per-lesion statistics, patients are resampled together with all their lesions (cluster bootstrap).

For the comparison with baselines, we additionally rank all methods with the significance-based ranking of the challenge: per metric, every pair of methods is compared with a one-sided Wilcoxon test (Holm-corrected), and a method scores (number of methods it significantly outperforms $+1$)/K for K methods. Scores are averaged into accuracy (CT Dice, HD95), PET preservation ($|\Delta\mathrm{MTV}|$, $|\Delta\mathrm{TLG}|$), and regularity (NDV, after subtracting a practical-equivalence threshold of 0.005 % per pair), and combined into $100 \cdot \mathrm{acc}^{0.4}\,\mathrm{PET}^{0.4}\,\mathrm{reg}^{0.2}$. The challenge additionally scores Dice and HD95 of organs with physiological PET uptake, for which no labels are available to us; we omit them. Dice or HD95 of lesions would not be meaningful either, because lesions appear, disappear, and change between examinations. The stability of the ranks is shown by 95 % rank intervals over 1000 patient bootstrap samples.

3.3 Tuning and Model Selection

All tuning used the validation set and the challenge ranking above as the selection rule: the candidate settings of one grid are ranked against each other, and the top-ranked one is kept.

- **IO:** on the network trained with both losses, step-wise: first learning rate $\times$ iterations ($\{0.0125, 0.025, 0.05, 0.1\} \times \{10, 20, 40\}$), then a common scale of the four PET weights ($\times\{0, 0.1, 0.25, 0.5, 1, 2, 4\}$ of the training-derived values). The ranking does not reward rigidity (its cost in Dice is significant, while folding stays below the threshold for every weight), so it would set λ_{rig} to zero; we fixed $\lambda_{\mathrm{rig}} = 0.25$ instead, which costs 0.3 Dice points on validation.

- **NiftyReg**:[28,29] `reg_aladin` followed by `reg_f3d` on the GPU with normalised mutual information (local NCC is not available on the GPU), over the bending-energy weights

$\{0.0002, 0.0005, 0.001, 0.005, 0.02\}$ and final control-point spacings of $\{3, 5, 7, 9\}$ voxels; chosen: 0.0002 and 9 voxels.

- **ConvexAdam**:[17] after the same ANTs affine as our method, over the regularisation weight $\lambda \in \{0.1, 0.25, 0.5, 1.25, 2, 4\}$; chosen: 0.1.

Several choices ended at the edge of their grid; extending the grids changed the validation scores only marginally.

Choice of the full method. Before the experiments, we defined our full method as the network trained with both losses followed by IO with both losses. To check this choice, we also ranked all variants of Sec. 3.4 on the validation set. This ranking favoured the variant without any of the losses, because accuracy differences between variants are significant while differences in MTV and TLG rarely are. We therefore report both in the comparison with baselines, together with the variant that uses the losses only in IO.

3.4 Experiments

Effect of the losses. Four networks were trained with the same alignment and regularisation terms and with neither, the PET-preservation, the rigidity, or both groups of losses. Each is evaluated as network output and with IO using the same losses as in its training. As a reference for early stopping, the network without the losses is also evaluated at the checkpoint with the best validation tumour preservation.

Losses in training or only in IO. The network trained without the losses is combined with IO using PET preservation, rigidity, or both, and compared with the networks trained with these losses

(and IO with the same losses). To measure what the network contributes, IO with both losses is also run directly after the affine pre-registration.

Contribution of each PET term. On the network trained without the losses, IO with all four PET terms is compared with IO in which one term at a time is switched off.

Comparison with baselines. Initial alignment, the affine pre-registration, NiftyReg, Convex-Adam, IO after the affine, and our variants are compared on the test set. Runtimes are per pair on one NVIDIA RTX A6000 GPU and include the affine step.

Rigidity penalties. On the network trained with the PET losses, IO with the PET terms is combined with no rigidity, Staring, Staring with erosion, or the per-bone rigid fit. Because the penalties have different units, a reference weight is first calibrated for each so that its initial value on five validation pairs equals that of the rigid fit at $\lambda_{\mathrm{rig}} = 0.25$ (Staring 0.66, Staring with erosion 1.03). Each method is then evaluated on the validation set at $\{0.1, 0.3, 1, 3, 10\}$ times its reference weight, giving a curve of CT Dice against within-bone distance distortion. For the test set, we choose for each method the largest weight whose validation Dice is within one Dice point of the variant without rigidity.

Accuracy–preservation trade-off. On the network trained without the losses, IO with the PET terms (without rigidity) is run with all four PET weights scaled by $\{0, 10^{-4}, 3{\cdot}10^{-4}, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3$ relative to the training-derived values; $\times 0.1$ is the setting chosen by the IO tuning. Besides the evaluation on the ground-truth labels, we compute CT Dice, $|\Delta\mathrm{MTV}|$, and $|\Delta\mathrm{TLG}|$ on the predicted labels that IO optimises, with the same definitions. To separate the effect of the predicted lesion

masks from that of the optimisation, IO is also run with the ground-truth lesion masks instead of the predicted ones (organ labels remain predicted), at the tuned and at the full PET weights.

Missed and detected lesions. Each ground-truth lesion of the moving scan is assigned to a group by the fraction of its voxels covered by the predicted lesion mask: missed (0 %), partly covered ($< 50\,\%$), or covered ($\geq 50\,\%$). For each group, we report two quantities, both computed per scan as in Eq. (4) and averaged over patients: the change within the group, using only the group's lesions, and the contribution of the group to the evaluation metric, i.e. the change if only the group's lesions changed and all other lesions kept their volume and uptake. Since changes of different lesions in a scan partly cancel, the contributions add up to slightly more than the evaluation metric.

4 Results

All results are on the test set (28 patients, 38 pairs); values are mean±SD over patients, and differences are given in percentage points.

4.1 Effect of the PET-Preservation and Rigidity Losses

Table 2 shows the four networks trained with neither, the PET-preservation, the rigidity, or both groups of losses, as network output and with IO using the same losses. Training with the PET losses lowered the volume change of the lesions from 10.6 % to 7.6 % and the uptake change from 10.8 % to 7.6 %, at the cost of 1.9 Dice points. IO raised the Dice of every network by 2.8 to 4.4 points and kept the effect of the losses: with the PET losses, $|\Delta\mathrm{MTV}|$ was 6.3 % instead of 10.2 % and $|\Delta\mathrm{TLG}|$ 7.2 % instead of 11.9 %, for 2.8 Dice points. The rigidity loss alone did not change the Dice of the network and cost 0.5 points with IO, and it lowered $|\Delta\mathrm{MTV}|$ in both cases (network 9.4 %, with IO 7.9 %). Combining both groups preserved about as well as the PET losses alone

Table 2 Effect of the PET and rigidity losses on the test set (28 patients, 38 pairs). Each network is trained with the given losses and evaluated as is (left) and with instance optimisation (IO) using the same losses (right). Mean±SD over patients. †/‡: significantly better/worse than "Neither" in the same block (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

	Network			Network + IO		
Losses	Dice (%) ↑	$\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓	$\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓	Dice (%) ↑	$\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓	$\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓
Neither	74.8±4.5	10.6±10.2	10.8±11.7	79.2±4.3	10.2±10.1	11.9±12.5
PET	72.9±4.4‡	7.6±9.0†	7.6±9.3†	76.4±4.7‡	6.3±9.1†	7.2±9.4†
Rigidity	74.8±4.5	9.4±9.5†	9.5±10.9†	78.7±4.2‡	7.9±9.0†	10.1±11.0
PET + rigidity	72.8±4.3‡	7.4±9.0†	7.7±10.1†	75.6±4.8‡	6.2±8.7†	7.9±9.0

and was the least accurate variant (72.8 % and 75.6 % Dice). HD95 followed the Dice in every case (network 7.2 to 7.5 mm, with IO 6.8 to 7.2 mm). No variant folded: the non-diffeomorphic volume of every pair stayed below the 0.005 % threshold of the challenge (at most 0.6 ppm on average).

Selecting the network without the losses at its checkpoint with the best validation tumour preservation, instead of the best accuracy, is no substitute for the losses: it cost 3.0 Dice points while $\lvert\Delta\mathrm{MTV}\rvert$ and $\lvert\Delta\mathrm{TLG}\rvert$ fell by only 0.4 and 0.9 points (not significant).

4.2 Losses in Training or Only in IO

Figure 2 compares, for each group of losses, the network trained with the losses (followed by IO with them) with the network trained without them, followed by the same IO. Using the losses only in IO was more accurate: Dice increased by 1.2 points [95 % CI 0.4, 1.9] for the PET losses and by 1.7 points [1.1, 2.2] for both groups (both significant), and was unchanged for rigidity. Preservation was not significantly worse: $\lvert\Delta\mathrm{MTV}\rvert$ increased by 1.3 [0.0, 2.6], 0.5 [0.0, 1.0], and 0.8 [-1.0, 2.7] points for PET, rigidity, and both, and $\lvert\Delta\mathrm{TLG}\rvert$ by at most 1.5 points. Without IO, in contrast, the training losses are the only means of preservation (Table 2, left).

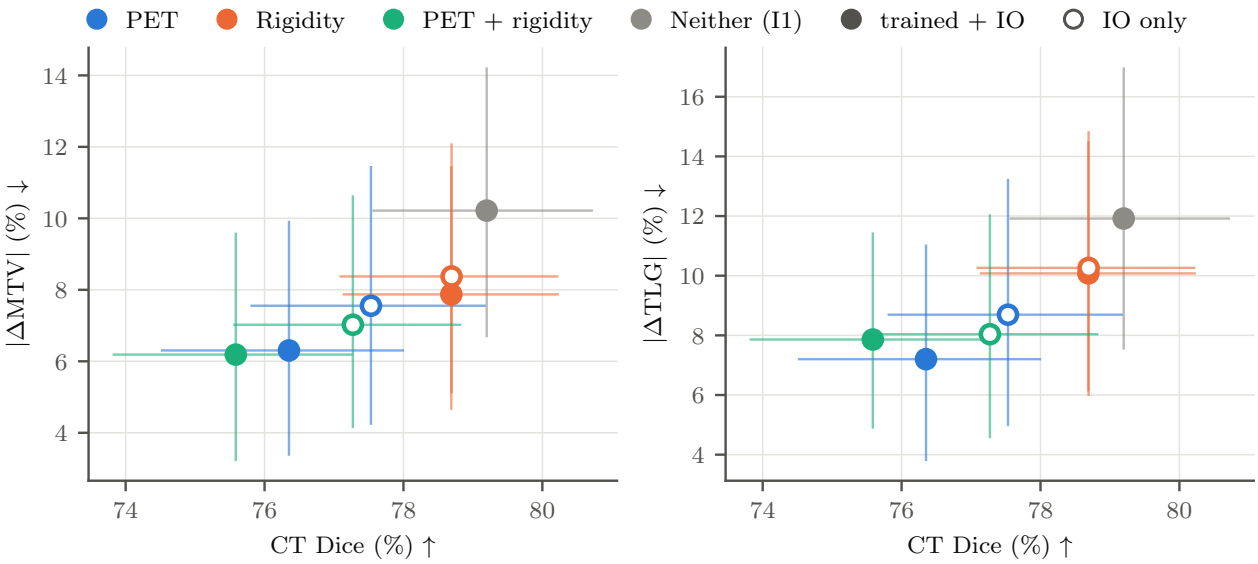


Fig 2 Losses in training and IO versus only in IO (test set, 28 patients). For each loss set, filled: network trained with the losses, then IO with them; hollow: network trained without them (T1), same IO. Grey: neither loss (I1). Points: patient means with 95 % patient-bootstrap confidence intervals.

4.3 Per-Lesion Volume Change

Figure 3 shows the volume change of the 1327 individual lesions (26 patients with lesions). Small lesions changed most in every condition: for bone-associated lesions with IO and without the losses, the median change was 33 % (small), 17 % (medium), and 9 % (large). A small lesion has fewer than about 20 voxels, so gaining or losing one or two voxels is already a large relative change. Without the losses, IO increased the distortion of lesions, most of all of small bone lesions (median from 20 % to 33 %); bone lesions tended to grow (median signed change $+3$ to $+10$ %) and soft-tissue lesions to shrink (-9 to -13 %). The rigidity loss acted only on bone-associated lesions, as expected from its construction: with IO, it lowered the median change of small, medium, and large bone lesions from 33 % to 20 %, 17 % to 12 %, and 9 % to 5 %, and left soft-tissue lesions unchanged (17 to 22 %). The PET losses acted in both locations, for example from 18 % to 10 % for medium soft-tissue lesions and from 33 % to 13 % for small bone lesions; only for small soft-tissue lesions, the smallest group (41 lesions), no clear effect was visible. With both groups, bone

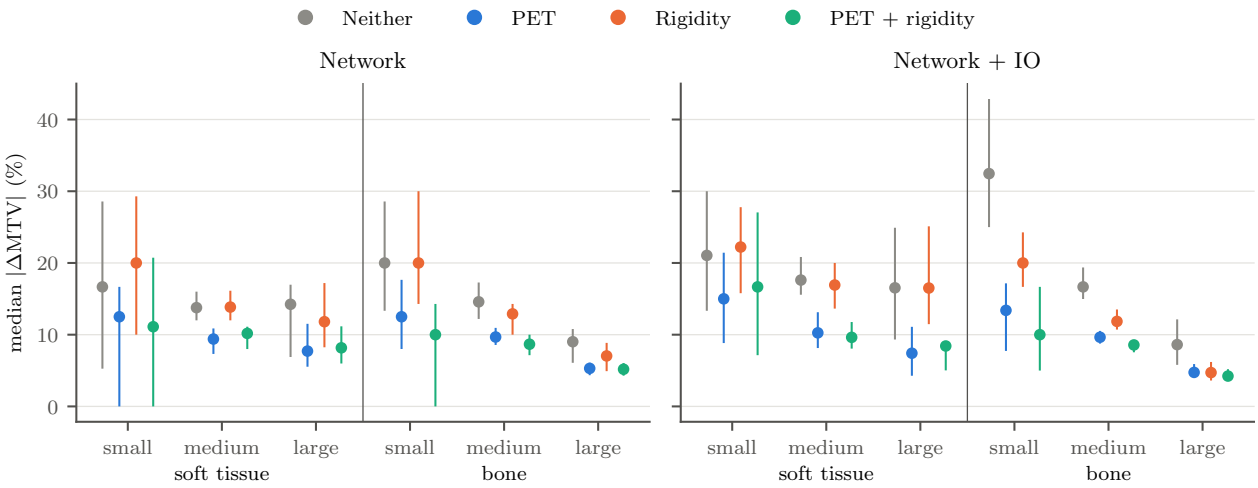


Fig 3 Per-lesion volume change on the test set (1327 lesions of 26 patients), by location (soft tissue / bone-associated, $\geq$50 % of the lesion on bone) and size (small $<$0.5 ml, medium $<$5 ml, large). Median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: with instance optimisation (IO) using the same losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap. Lesions below 5 voxels are excluded.

lesions changed least at every size (10 %, 9 %, and 4 %), and their systematic growth disappeared (median signed change about 0). The medians of small lesions take only a few distinct values (e.g. 10.0 %, 12.5 %, 20.0 %) because they are ratios of small voxel counts.

4.4 Contribution of Each PET Term

Table 3 switches off one PET term at a time in IO on the network trained without the losses. The two Jacobian terms carried the volume preservation: without $\mathcal{L}_{\overline{J}}$, $|\Delta\mathrm{MTV}|$ increased from 7.6 % to 8.3 %, and without $\mathcal{L}_J$ to 7.8 % (both significant). The volume term $\mathcal{L}_{\mathrm{MTV}}$ mainly cost accuracy: without it, Dice increased by 1.3 points (from 77.5 % to 78.8 %, significant), while $|\Delta\mathrm{MTV}|$ increased only from 7.6 % to 8.0 % and $|\Delta\mathrm{TLG}|$ decreased from 8.7 % to 8.2 % (neither significant). Switching off $\mathcal{L}_{\mathrm{TLG}}$ changed no metric significantly ($|\Delta\mathrm{TLG}|$ 9.1 %, $|\Delta\mathrm{MTV}|$ 7.2 %). No single term changed $|\Delta\mathrm{TLG}|$ significantly, and all changes of $|\Delta\mathrm{MTV}|$ were below one percentage point.

Table 3 Contribution of each PET term in IO (test set, 28 patients): network trained without PET losses (T1), IO with all PET terms or with one term removed. Mean±SD over patients. †/‡: significantly better/worse than with all terms (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

IO PET terms	Dice (%) ↑	$\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓	$\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓
All PET terms	77.5±4.5	7.6±10.0	8.7±11.1
without $\mathcal{L}_{\mathrm{MTV}}$	78.8±4.3†	8.0±9.8	8.2±11.0
without $\mathcal{L}_{\overline{J}}$	77.6±4.6	8.3±9.9‡	8.9±11.2
without $\mathcal{L}_{\mathrm{TLG}}$	77.5±4.5	7.2±10.1	9.1±10.8
without $\mathcal{L}_{J}$	77.6±4.5†	7.8±9.9‡	9.1±10.9

4.5 Comparison with Baselines

Table 4 and Fig. 4 compare our variants with the baselines. Every variant of ours was significantly more accurate than NiftyReg (69.4 % Dice, 8.6 mm HD95), ConvexAdam (61.7 %, 9.2 mm), and the affine pre-registration (46.4 %, 13.9 mm), and none of them folded. NiftyReg, tuned on the validation set where it rarely folded, exceeded the folding threshold on 21 % of the test pairs (mean 1170 ppm, driven by three pairs with 0.8 to 1.3 %). IO directly after the affine pre-registration reached only 52.7 % Dice: within the same iteration budget, IO cannot replace the network. Its MTV and TLG change was the lowest of all methods (3.4 %), because the deformation it adds to the affine is small.

Among our variants, IO without the PET and rigidity losses was the most accurate (79.2 % Dice), and our full method, with the losses in training and IO, preserved best ($|\Delta\mathrm{MTV}|$ 6.2 %, significantly lower than without the losses). The variant with the losses only in IO ranked first (77.3 % Dice, 7.0 % $|\Delta\mathrm{MTV}|$), followed by the full method and the variant without the losses; their rank intervals overlap (1–5). The ranking was decided mainly by accuracy: after correction for multiple comparisons, few differences in preservation were significant (on $|\Delta\mathrm{MTV}|$, each variant with the losses outperformed one other method and IO after the affine two; on $|\Delta\mathrm{TLG}|$, only IO after the affine outperformed others), and none in folding. The network alone needs 11 s per pair, as fast as ConvexAdam and ten times faster than NiftyReg (108 s); IO adds about 2.5 minutes. Figure 5

Table 4 Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs). Values are mean±SD over patients (a patient's pairs are averaged first). Rank: challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimisation (IO); IO "–": no IO, "basic": IO with the similarity and regularisation terms only. †/‡: Ours + IO with PET + rig. in training and IO (our full method) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). > thr.: pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000, including the affine step. Affine + IO: our instance optimisation directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.

Method	Training	IO	Rank ↓	Dice (%) ↑	HD95 (mm) ↓	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓	NDV (ppm) ↓	> thr. (%)	Runtime (s)
Initial	–	–	–	9.5±8.1†	44.37±21.32†	0.0±0.0	0.0±0.0	0±0	0	–
Affine	–	–	9 [8–9]	46.4±10.8†	13.92±4.74†	4.6±4.6	5.7±3.9	**0**±0	0	10.4±0.3
NiftyReg	–	–	7 [4–7]	69.4±4.7†	8.60±3.69†	7.6±12.5	11.3±13.7	1170±3120	21	107.7±24.9
ConvexAdam	–	–	8 [7–8]	61.7±4.5†	9.16±3.14†	9.3±12.0	9.6±15.8	**0**±0	0	11.3±0.6
Affine + IO	–	PET + rig.	5 [1–8]	52.7±12.1†	13.26±4.77†	**3.4**±3.2	**3.4**±3.6	0.1±0.099	0	155.5±2.6
Ours	–	–	6 [3–7]	74.8±4.5†	7.23±2.32	10.6±10.2†	10.8±11.7	0.0037±0.01	0	11.0±0.4
Ours	PET + rig.	–	4 [1–6]	72.8±4.3†	7.46±2.41†	7.4±9.0	7.7±10.1	0.0055±0.012	0	11.3±0.3
Ours + IO	–	basic	2 [1–5]	**79.2**±4.3‡	**6.78**±2.30‡	10.2±10.1†	11.9±12.5	0.58±0.26	0	153.9±2.6
Ours + IO	–	PET + rig.	1 [1–5]	77.3±4.5‡	7.01±2.33‡	7.0±8.8	8.0±10.2	0.25±0.25	0	157.2±2.6
Ours + IO	PET + rig.	PET + rig.	2 [1–5]	75.6±4.8	7.24±2.38	6.2±8.7	7.9±9.0	0.084±0.089	0	156.6±2.7

shows a test pair whose lesion volume grew tenfold between baseline and follow-up (from 2400 to 24 191 voxels). The scans were initially far apart (CT Dice 3.5 %); our full method aligned them to 81.0 % Dice (NiftyReg 69.5 %, ConvexAdam 65.6 %), while the volume and uptake of the 163 follow-up lesions changed by only $-0.1\,\%$ and $+0.9\,\%$ and no voxel folded. Before registration, the CT overlay (Fig. 5c) and the label residual (e) show misalignment across the whole trunk; after registration (d, f), the residual is reduced to thin bands at organ boundaries. [TODO: Once the figure with IO is final: where the label residual remains, what $\det J$ shows.]

4.6 Rigidity Penalties

[TODO: After the validation sweeps and test runs: curve of Dice vs. distance distortion (validation), test table with bone/soft-tissue alignment, distance distortion, NDV in bone and boundary band, stencil leakage, erosion retention. Leakage on all 560 scans: 21.2 % of stencil reads leave the bone; erosion keeps 35.5 % of bone voxels and removes 23 % of bones (ribs: 2.4 % kept, 58 %

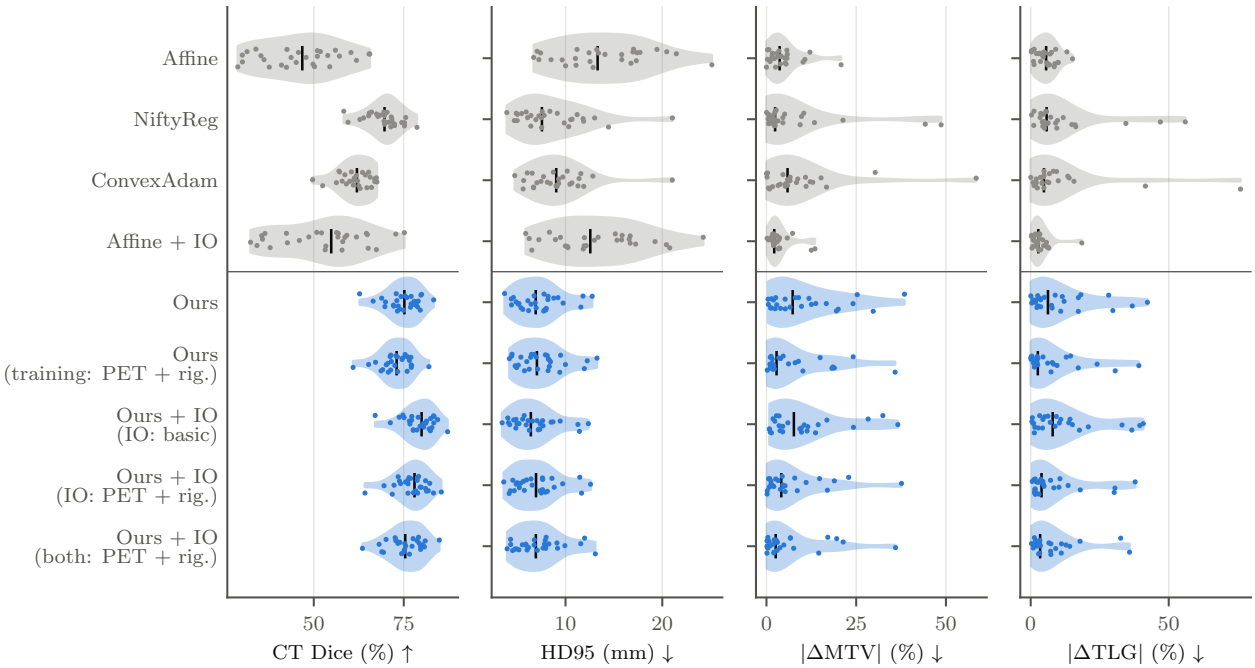


Fig 4 Patient-level results on the test set (28 patients; a patient's pairs are averaged). Violins: kernel density over patients, line: median; each dot is one patient. In brackets: where the PET and rigidity losses (PET + rig.) act; basic: IO with the similarity and regularisation terms only; both: training and IO. Initial alignment is omitted (CT Dice 9.5 %, HD95 44.4 mm; MTV/TLG change 0 by construction, Table 4).

removed).]

4.7 Accuracy–Preservation Trade-off

[TODO: Text for the trade-off once the runs at $\times\{0, 10^{-4}, 3\cdot10^{-4}, 0.001, 0.003\}$ and with ground-truth lesion masks are done.]

4.8 Missed and Detected Lesions

Figure 7 groups the ground-truth lesions by how much of each the predicted lesion mask covers: 214 lesions were missed (6 % of the lesion volume), 155 partly covered (6 %), and 1066 covered (88 %).

PET IO reduced the change only of covered lesions, from 7.8 % to 4.9 % for $|\Delta\mathrm{MTV}|$ and from 7.9 % to 5.1 % for $|\Delta\mathrm{TLG}|$ (IO without and with the PET losses at the tuned weights). Missed and

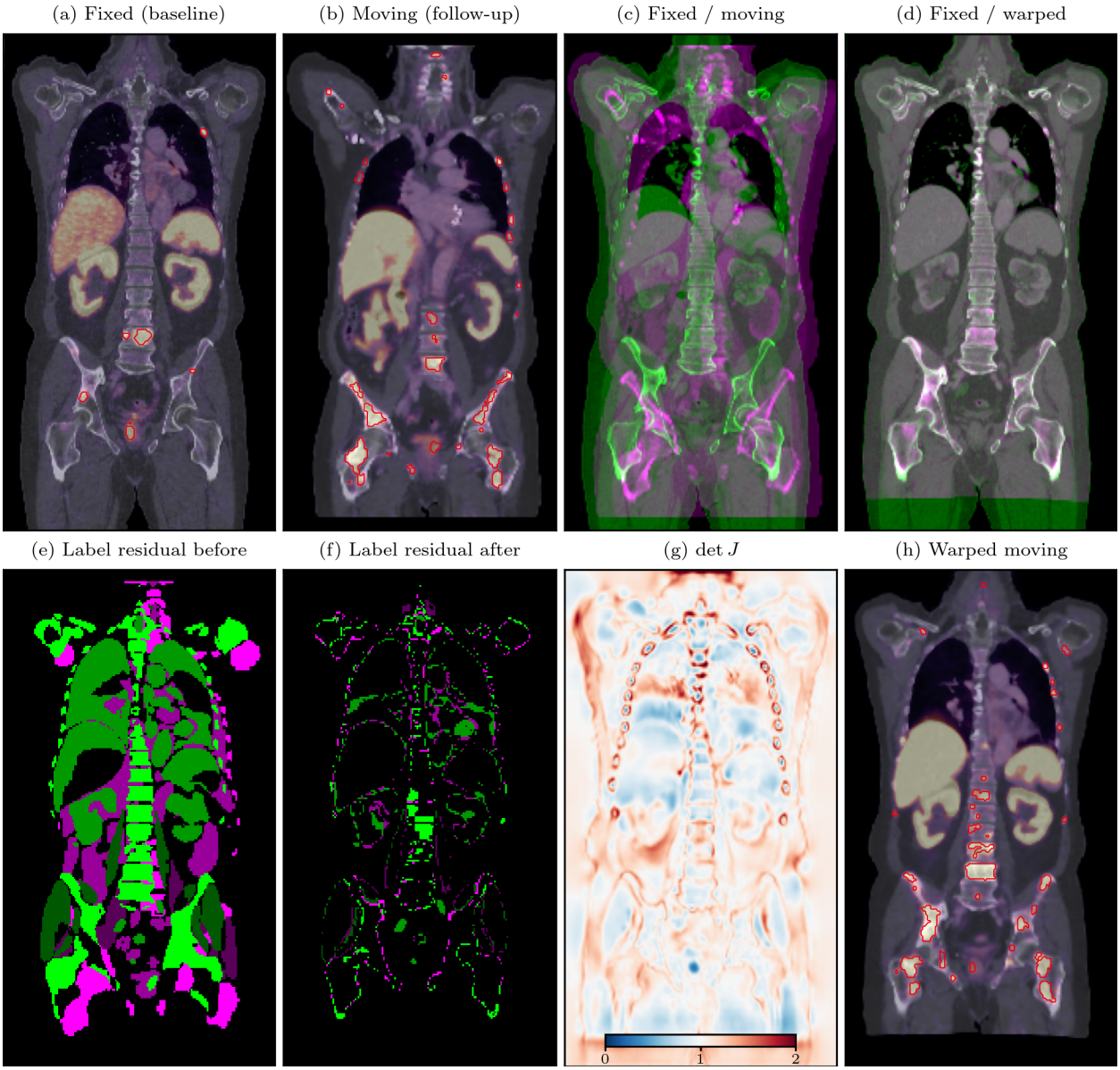


Fig 5 Test pair with a large change in tumour volume (patient 0146; lesion volume 2400 voxels at baseline, 24191 at follow-up), registered by our network (PET and rigidity losses in training, no IO); one coronal slice. (a) Fixed and (b) moving CT with PET (SUV 0–8, 35 % opacity) and ground-truth lesion contours (red). (c, d) CT of the fixed scan (green) overlaid with the moving scan (magenta) before (c) and after (d) registration; grey where they agree. (e, f) Organ labels that differ before (e) and after (f) registration: green where the fixed scan has a label that the moving scan lacks there, magenta where only the moving scan is labelled; bright, medium, and dark shades for bones, organs, and muscles. (g) Jacobian determinant of the transformation. (h) Warped moving CT with PET and warped lesion contours.

partly covered lesions changed as much with the PET losses as without them (missed: 14.9 % and 15.7 %; partly covered: 11.3 % and 11.5 %), since IO does not see them. After PET IO, missed lesions therefore contributed most of the remaining change: 4.8 of 7.6 points of $|\Delta\mathrm{MTV}|$ and 5.7

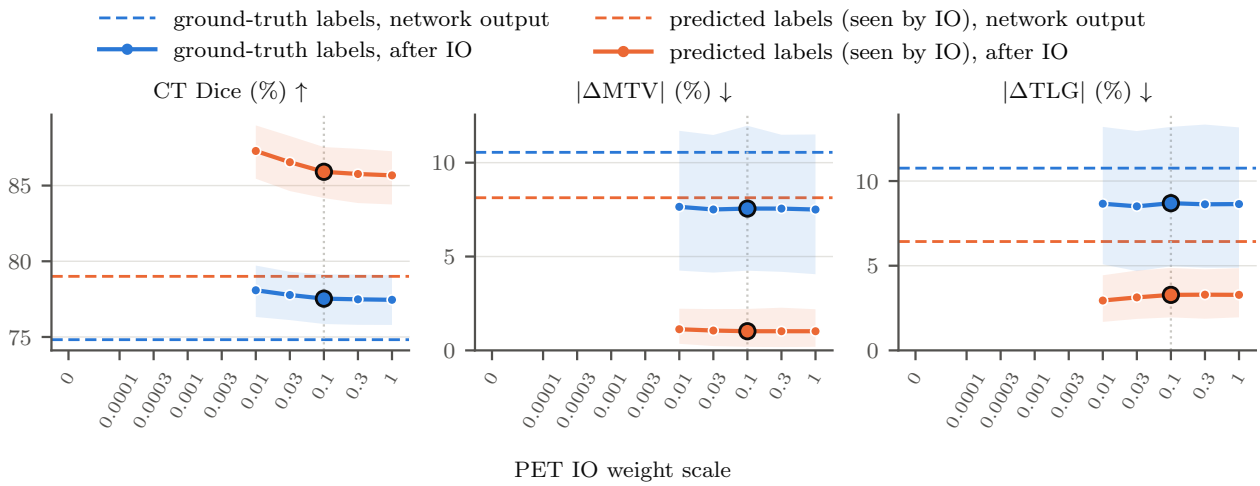


Fig 6 Effect of the PET loss weight in IO (test set): network trained without PET losses (T1) followed by IO with all PET weights scaled by the given factor (0: PET losses off; 1: the PET weights of network training). Each metric on the predicted labels that IO optimises (orange; organs: TotalSegmentator, lesions: nnU-Net) and on the ground-truth labels used for evaluation (blue). Dice over all 28 patients, lesion volume and uptake change over the 26 patients with lesions in both label sets. Solid: after IO; dashed: network output before IO, the same for every factor. Outlined point and dotted line: operating point tuned on the validation set. Points: patient means; bands: 95 % patient-bootstrap confidence intervals.

of 8.7 points of $|\Delta\mathrm{TLG}|$, although they make up only 6 % of the lesion volume. Covered lesions contributed 2.6 and 3.2 points. Scaling the PET weights from $\times 0.1$ to $\times 1$ changed none of these values by more than 0.2 points.

Without the PET losses, IO increased the change of covered lesions over the network output, from 7.4 % to 7.8 % ($|\Delta\mathrm{MTV}|$) and from 6.9 % to 7.9 % ($|\Delta\mathrm{TLG}|$), and the overall $|\Delta\mathrm{TLG}|$ from 10.8 % to 11.9 %; the overall $|\Delta\mathrm{MTV}|$ fell slightly, from 10.6 % to 10.2 %, because missed and partly covered lesions changed less. This agrees with the per-lesion analysis (Sec. 4.3), in which IO without the losses increased the volume change of lesions. [TODO: Covered lesions still contribute 2.6 points of $|\Delta\mathrm{MTV}|$ after PET IO, although IO sees them: add the result of IO with ground-truth lesion masks, which tests whether this part is due to the inaccuracy of the predicted masks.]

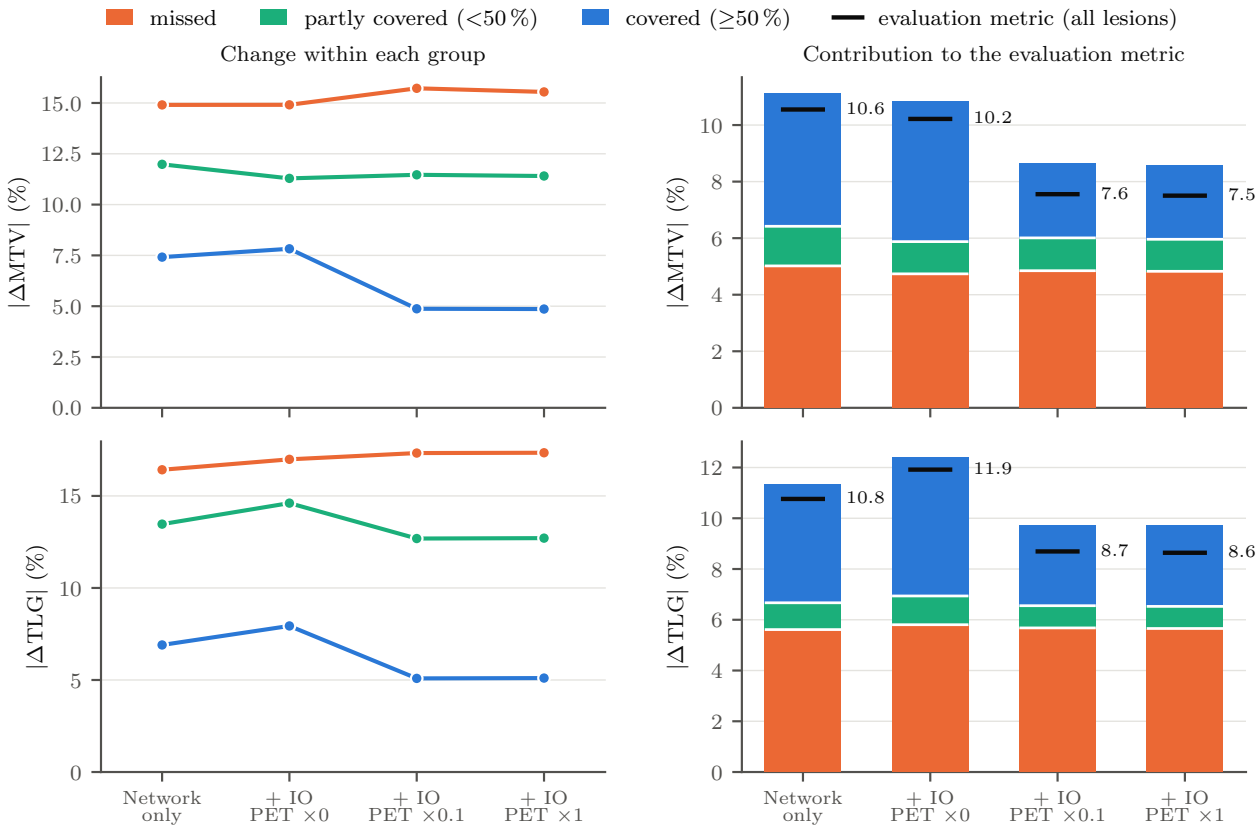


Fig 7 Lesion volume and uptake change of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers: missed (214 lesions, 6 % of volume), partly covered (155 lesions, 6 % of volume), covered (1066 lesions, 88 % of volume). Test set; network trained without PET losses, IO with the PET weights scaled by the given factor. Left: change within each group (per scan, the change of the summed volume or uptake of the group's lesions; mean over the patients with lesions in the group). Right: contribution of each group to the evaluation metric (the change per scan if only the group's lesions changed; mean over patients); black: the evaluation metric, below the stacked total because changes of different lesions in a scan partly cancel.

5 Discussion

The PET-preservation losses reduced the change of lesion volume and uptake caused by registration, both in the network output and after IO, at a cost of about two to three Dice points; the rigidity loss reduced the volume change at a smaller cost. With IO, applying the losses only in IO preserved about as well as training with them and was more accurate. All our variants were more accurate than tuned NiftyReg and ConvexAdam and did not fold.

Losses in training or only in IO. Which placement to choose depends on the time budget. If IO is affordable, about 2.5 minutes per pair on our GPU, the network can be trained for accuracy alone and the losses applied in IO: compared with training and IO with the losses, this gained 1.7 Dice points without significantly worse preservation. Without IO, the network alone registers a pair in 11 s, and training with the losses is then the only way to preserve lesions; selecting the checkpoint of the network without the losses for tumour preservation instead cost more accuracy and preserved hardly better. The training losses are therefore not superseded by IO, but they serve a different setting, e.g. fast registration of many follow-up examinations.

What limits preservation in IO. On the predicted lesion masks that IO optimises, the PET losses removed almost all volume change (from 8.1 % to 1.0 %), and weights ten times higher than the tuned ones changed none of the metrics. The remaining change on the ground-truth lesions is therefore not a matter of the optimisation or the weights but of the masks: lesions that nnU-Net missed cannot be preserved and contributed most of the remaining volume and uptake change, although they make up only 6 % of the lesion volume. Their weight comes from patients with little tumour volume: the metric is relative per scan and averaged over patients, and in the third of patients with the smallest lesion volume, missed lesions made up 45 % of the volume (18 % and 6 % in the other thirds) and caused 70 % of the missed-lesion contribution. Detecting more lesions at the price of precision is no remedy: on the validation set, IO with a recall-oriented mask (SUV threshold minus physiological organs, combined with nnU-Net), which detected more lesions but had a precision of 0.18, preserved worse than without the PET losses, because preserving the false-positive regions came at the expense of the true lesions. Better lesion segmentation, rather than a different loss weighting, is therefore the most promising way to improve preservation in

IO. [TODO: Add the runs with ground-truth lesion masks: how much of the remaining change disappears, including the part of covered lesions (inaccurate extent of detected lesions).]

Which PET terms matter. In IO, the two Jacobian terms carried the volume preservation, whereas the volume term $\mathcal{L}_{\mathrm{MTV}}$ cost 1.3 Dice points and contributed no significant preservation, and the uptake term $\mathcal{L}_{\mathrm{TLG}}$ had no significant effect. A smaller set of terms would likely suffice; we kept all four because they were chosen before the test set was evaluated, and all effects of single terms were below one percentage point.

Which lesions are affected. Aggregate MTV and TLG changes hide how individual lesions are deformed. Small lesions changed most in every condition, partly because a change of one or two voxels is already a large relative change. Without the losses, IO increased the distortion of lesions, most of all of small bone-associated lesions, and bone lesions tended to grow. The rigidity loss acted only on bone-associated lesions, as intended, and the PET losses on both bone and soft-tissue lesions; with both groups of losses, bone lesions changed least, and their systematic growth disappeared.

Evaluation and ranking. The significance-based ranking of the challenge was decided mainly by accuracy: differences in alignment between methods are consistent across patients and reach significance, whereas differences in MTV and TLG vary strongly between patients and rarely do. On the validation set, the ranking consequently chose our variant without any of the losses. The ranking also rewards methods that hardly deform: IO directly after the affine pre-registration had the lowest MTV and TLG change of all methods and ranked fifth with 52.7 % Dice, above our network without IO with 74.8 %. Preservation metrics should therefore be read together with ac-

curacy, not on their own. A further effect of the scorer is that warping the PET image changes TLG even for the identity transformation (2.9 % on average); this floor applies equally to all methods but limits how small a TLG change can be reported. Per-lesion changes, which do not cancel between lesions, would complement the aggregate metrics.

Registration limited by segmentation. Both the accuracy and the preservation of IO are bounded by the labels it uses. On the predicted organ labels, IO raised Dice from 79.0 % to 85.9 %, on the ground-truth labels only from 74.8 % to 77.5 %; for lesions, it preserved the predicted masks almost perfectly but not the missed lesions. Within a challenge whose metrics are computed on segmentations, better segmentation is thus a direct route to better registration scores, for organs as well as for lesions. Outside this setting, label overlap and the change of MTV and TLG are proxies: what matters clinically is whether registration supports downstream tasks such as lesion tracking, response assessment, or dose accumulation. Future work should therefore evaluate registration on such tasks and clinical endpoints rather than optimise further for segmentation-based metrics.

Baselines. NiftyReg, tuned on the validation set where it rarely folded, exceeded the folding threshold on 21 % of the test pairs, with three pairs above 0.8 %. The ranking selected its least-regularised setting because accuracy separated the settings more than folding; a setting chosen on the validation set did not transfer to new patients. Our networks with and without IO stayed below the threshold on every pair.

Rigidity penalties. [TODO: After the rigidity experiments: stencil leakage of finite-difference penalties at bone boundaries (21 % of stencil reads leave the bone), erosion removes most of the

skeleton at this resolution (35.5 % of bone voxels kept, ribs 2.4 %), per-bone rigid fit avoids both; results of the comparison.]

Limitations. The study uses a single cohort with 28 test patients, so small differences, in particular in MTV and TLG, could not be resolved, and generalisation to other scanners and tracers is untested. The images were resampled to about $2.7 \times 2.7 \times 3.3$ mm, at which many lesions consist of few voxels and small lesions are dominated by discretisation. The organ labels used for evaluation are themselves automatic (TotalSegmentator), and the lesion labels used by IO are predictions with a Dice of about 0.5, which, as shown above, limits preservation in IO. All patients, including the test patients, were part of the challenge data with which the earlier version of the method was developed; the test patients were not used for any decision in the present study. We studied one network architecture, and we measured the change caused by registration, not its effect on clinical decisions such as response assessment.

6 Conclusion

Losses that preserve lesion volume and uptake and keep bones rigid reduce the change of PET biomarkers caused by deformable registration of longitudinal whole-body PSMA PET/CT, at a moderate cost in alignment accuracy. When instance optimisation is affordable, the losses are best applied there, on a network trained for accuracy; without it, training with the losses is the only way to preserve lesions. In instance optimisation, the remaining change stems mainly from lesions that the predicted lesion masks miss, so better lesion segmentation, not stronger weighting, is the way to further preservation.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code, Data, and Materials Availability

All experiments use the publicly available training data of the Learn2Reg 2026 PSMAReg challenge,[3,4] which is adapted from the PSMA-PET-CT-Lesions dataset in The Cancer Imaging Archive[5] (DOI: 10.7937/r7ep-3x37). The private test data of the challenge were not used. The patient-level split into training, validation, and test sets is provided with the code. [TODO: Code repository link (training, IO, evaluation, split, and trained weights).]

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reviewed and take responsibility for all content. The use of AI tools for code generation is described in Sec. 2.5.

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List of Figures

1 Registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT scans are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimisation (IO) refines this deformation per pair before the moving scan is resampled into the fixed frame. Network training and IO use the same groups of loss terms: alignment, PET preservation, and deformation regularisation (Sec. 2.3). [TODO: Update the figure if the rigidity term or the IO-only variant should appear.]

2 Losses in training and IO versus only in IO (test set, 28 patients). For each loss set, filled: network trained with the losses, then IO with them; hollow: network trained without them (T1), same IO. Grey: neither loss (I1). Points: patient means with 95 % patient-bootstrap confidence intervals.

3 Per-lesion volume change on the test set (1327 lesions of 26 patients), by location (soft tissue / bone-associated, $\geq$50 % of the lesion on bone) and size (small $<$0.5 ml, medium $<$5 ml, large). Median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: with instance optimisation (IO) using the same losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap. Lesions below 5 voxels are excluded.

4 Patient-level results on the test set (28 patients; a patient's pairs are averaged). Violins: kernel density over patients, line: median; each dot is one patient. In brackets: where the PET and rigidity losses (PET + rig.) act; basic: IO with the similarity and regularisation terms only; both: training and IO. Initial alignment is omitted (CT Dice 9.5 %, HD95 44.4 mm; MTV/TLG change 0 by construction, Table 4).

5 Test pair with a large change in tumour volume (patient 0146; lesion volume 2400 voxels at baseline, 24191 at follow-up), registered by our network (PET and rigidity losses in training, no IO); one coronal slice. (a) Fixed and (b) moving CT with PET (SUV 0–8, 35 % opacity) and ground-truth lesion contours (red). (c, d) CT of the fixed scan (green) overlaid with the moving scan (magenta) before (c) and after (d) registration; grey where they agree. (e, f) Organ labels that differ before (e) and after (f) registration: green where the fixed scan has a label that the moving scan lacks there, magenta where only the moving scan is labelled; bright, medium, and dark shades for bones, organs, and muscles. (g) Jacobian determinant of the transformation. (h) Warped moving CT with PET and warped lesion contours.

6 Effect of the PET loss weight in IO (test set): network trained without PET losses (T1) followed by IO with all PET weights scaled by the given factor (0: PET losses off; 1: the PET weights of network training). Each metric on the predicted labels that IO optimises (orange; organs: TotalSegmentator, lesions: nnU-Net) and on the ground-truth labels used for evaluation (blue). Dice over all 28 patients, lesion volume and uptake change over the 26 patients with lesions in both label sets. Solid: after IO; dashed: network output before IO, the same for every factor. Outlined point and dotted line: operating point tuned on the validation set. Points: patient means; bands: 95 % patient-bootstrap confidence intervals.

7 Lesion volume and uptake change of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers: missed (214 lesions, 6 % of volume), partly covered (155 lesions, 6 % of volume), covered (1066 lesions, 88 % of volume). Test set; network trained without PET losses, IO with the PET weights scaled by the given factor. Left: change within each group (per scan, the change of the summed volume or uptake of the group's lesions; mean over the patients with lesions in the group). Right: contribution of each group to the evaluation metric (the change per scan if only the group's lesions changed; mean over patients); black: the evaluation metric, below the stacked total because changes of different lesions in a scan partly cancel.

List of Tables

1 Loss weights of Eq. (1) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimisation (IO).

2 Effect of the PET and rigidity losses on the test set (28 patients, 38 pairs). Each network is trained with the given losses and evaluated as is (left) and with instance optimisation (IO) using the same losses (right). Mean±SD over patients. †/‡: significantly better/worse than "Neither" in the same block (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

3 Contribution of each PET term in IO (test set, 28 patients): network trained without PET losses (T1), IO with all PET terms or with one term removed. Mean±SD over patients. †/‡: significantly better/worse than with all terms (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

4 Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs). Values are mean±SD over patients (a patient's pairs are averaged first). Rank: challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimisation (IO); IO "–": no IO, "basic": IO with the similarity and regularisation terms only. †/‡: Ours + IO with PET + rig. in training and IO (our full method) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). $>$ thr.: pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000, including the affine step. Affine + IO: our instance optimisation directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.